No.	Status	Date
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UT Thickness Survey - ATU-100 Rich Amine (Pre-MOC)

Scope and Objectives

Larkhall NDE Services Inc. was engaged to perform a ultrasonic thickness survey on the rich amine piping within the ATU-100 at the Hamilton Plant. This inspection serves as the baseline for a Management of Change (MOC) process detailed in QSCP-DMR-2014-002, which addresses a DEA strength increase. The primary objective is to establish current wall thicknesses for lines 4-RA-101, 4-RA-102, 6-RA-103, and 4-RA-104 prior to the expected increase in acid-gas loading and associated corrosion rates.

The survey follows the requirements of QSCP-IP-ATU100 Rev 2, which classifies these rich amine circuits as requiring a 3-year thickness interval. This inspection is the first job performed by Larkhall NDE Services Inc. at this facility, following the previous 5-year Class 1 survey documented in HIS-2011-0207. The scope includes the addition of new CMLs downstream of the let-down valve as mandated by the recent damage mechanism review.

Personnel and Responsibilities

The field inspection was conducted by Marcus Chen, acting as the lead UT technician with qualifications meeting ASNT SNT-TC-1A UT Level II. He was responsible for site data acquisition, CML identification, and technical oversight of the field team. Trevor Nolan assisted in the survey as a UT technician with ASNT SNT-TC-1A UT Level I (2014) certification, performing surface preparation and secondary readings under direct supervision.

Technical review and quality assurance of this report and the underlying data were performed by David Park, serving as the Level III / Quality Manager. All activities were executed in accordance with the Hamilton Plant asset integrity requirements and the specific NDE guidelines established for the ATU-100 unit.

Methodology and Equipment

Ultrasonic thickness measurements were performed under QSCP-NDT-001 Rev 2, utilizing digital pulse-echo techniques. The procedure incorporates requirements for CUI program management, including insulation removal and reinstatement where profile radiography was not utilized. The inspection adhered to the technical standards set forth in ASTM E797 and API 570 for the in-service inspection of piping systems.

The instrument used was an Olympus 37DL PLUS 0611452, which was verified against LNDE-CAL-2014-0655. Field calibration was performed using LNDE-CS-001, a 4-step carbon steel calibration block with steps ranging from 0.100 to 0.400 in. All measurements on the A106 Gr.B carbon steel piping were recorded to a precision of 0.01 mm to ensure accurate short-term corrosion rate calculations.

Inspection Criteria and Design Limits

Evaluation of the piping systems was conducted against the design limits specified in ASME B31.3. For lines 4-RA-101, 4-RA-102, and 4-RA-104, the nominal wall thickness is 8.56 mm with a retirement thickness (t-min) of 3.81 mm. The 6-RA-103 rich amine header has a nominal thickness of 10.97 mm and a t-min of 4.57 mm. Design conditions for these lines are 862 kPa(g) with temperatures ranging from 82 °C to 127 °C.

Per site rules, the retirement thickness is defined as the line's t-min. Corrosion rates are determined by selecting the larger of the long-term rate or the short-term rate, provided the readings are at least 3 years apart. As the last rich-amine readings were recorded in 2011-02, this survey provides the necessary 3-year data point to calculate the current short-term corrosion rate.

Summary of Results

Data obtained during this survey were compared against the historical values in HIS-2011-0207. The rich amine circuits remain on a 3-year thickness interval due to the higher acid-gas loading expected from the MOC. All readings for 4-RA-101, 4-RA-102, 6-RA-103, and 4-RA-104 were found to be above the established t-min values. No Non-Conformance Reports (NCRs) were issued during this inspection cycle.

Remaining life calculations were performed using the formula (t - t-min) / rate. In accordance with site rules, the next thickness reading must be scheduled at no more than half the remaining life. The CUI program requirements from QSCP-IP-ATU100 Rev 2 were also applied to the insulated sections of the rich amine lines operating within the CUI range, ensuring compliance with the 5-year window for insulation inspection.

Thickness readings, measured 2014-09-19

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2011-02-18	8.32	8.07	0.024	0.070	>50	2017-09-19	
4-RA-101-02	8.56	3.81	2011-02-18	8.20	8.02	0.028	0.050	>50	2017-09-19	
4-RA-101-03	8.56	3.81	2011-02-18	7.98	7.88	0.039	0.028	>50	2017-09-19	
4-RA-101-04	8.56	3.81	2011-02-18	8.21	8.18	0.024	0.008	>50	2017-09-19	
4-RA-101-05	8.56	3.81	2011-02-18	8.16	7.98	0.033	0.050	>50	2017-09-19	
4-RA-101-06	8.56	3.81	—	—	7.91	—	—	—	2017-09-19	
4-RA-101-07	8.56	3.81	2011-02-18	8.30	8.22	0.021	0.022	>50	2017-09-19	
4-RA-101-08	8.56	3.81	2011-02-18	8.02	7.71	0.052	0.086	45.1	2017-09-19	
4-RA-101-09	8.56	3.81	2011-02-18	8.32	8.27	0.016	0.014	>50	2017-09-19	
4-RA-101-10	8.56	3.81	2011-02-18	8.28	8.21	0.025	0.020	>50	2017-09-19	
4-RA-101-11	8.56	3.81	2011-02-18	8.04	7.88	0.043	0.045	>50	2017-09-19	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2011-02-18	8.33	8.25	0.016	0.022	>50	2017-09-19	
4-RA-102-02	8.56	3.81	2011-02-18	7.99	7.89	0.037	0.028	>50	2017-09-19	
4-RA-102-03	8.56	3.81	2011-02-18	8.04	7.95	0.033	0.025	>50	2017-09-19	
4-RA-102-04	8.56	3.81	2011-02-18	7.96	7.93	0.033	0.008	>50	2017-09-19	
4-RA-102-05	8.56	3.81	2011-02-18	8.07	7.84	0.038	0.064	>50	2017-09-19	
4-RA-102-06	8.56	3.81	2011-02-18	8.01	7.95	0.030	0.017	>50	2017-09-19	
4-RA-102-07	8.56	3.81	2011-02-18	8.10	7.98	0.032	0.033	>50	2017-09-19	
4-RA-102-08	8.56	3.81	2011-02-18	7.88	7.58	0.049	0.084	45.0	2017-09-19	
4-RA-102-09	8.56	3.81	2011-02-18	7.98	7.84	0.038	0.039	>50	2017-09-19	
4-RA-102-11	8.56	3.81	2011-02-18	8.19	8.12	0.020	0.020	>50	2017-09-19	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2011-02-18	10.47	10.62	0.021	-0.042	>50	2017-09-19	
6-RA-103-02	10.97	4.57	2011-02-18	10.23	10.06	0.056	0.047	>50	2017-09-19	
6-RA-103-03	10.97	4.57	2011-02-18	10.42	10.39	0.032	0.008	>50	2017-09-19	
6-RA-103-04	10.97	4.57	2011-02-18	9.97	9.79	0.064	0.050	>50	2017-09-19	
6-RA-103-05	10.97	4.57	2011-02-18	10.10	9.81	0.062	0.081	>50	2017-09-19	
6-RA-103-06	10.97	4.57	2011-02-18	10.56	10.43	0.031	0.036	>50	2017-09-19	
6-RA-103-07	10.97	4.57	2011-02-18	10.55	10.37	0.033	0.050	>50	2017-09-19	
6-RA-103-08	10.97	4.57	2011-02-18	10.44	10.49	0.026	-0.014	>50	2017-09-19	
6-RA-103-09	10.97	4.57	2011-02-18	10.56	10.39	0.033	0.047	>50	2017-09-19	
6-RA-103-10	10.97	4.57	2011-02-18	10.05	9.93	0.059	0.033	>50	2017-09-19	
6-RA-103-11	10.97	4.57	—	—	10.10	—	—	—	2017-09-19	
6-RA-103-13	10.97	4.57	2011-02-18	10.45	10.38	0.020	0.020	>50	2017-09-19	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2011-02-18	7.95	7.84	0.036	0.031	>50	2017-09-19	
4-RA-104-02	8.56	3.81	2011-02-18	8.14	8.09	0.025	0.014	>50	2017-09-19	
4-RA-104-03	8.56	3.81	2011-02-18	8.56	8.46	0.032	0.028	>50	2017-09-19	
4-RA-104-04	8.56	3.81	2011-02-18	8.60	8.49	0.007	0.031	>50	2017-09-19	
4-RA-104-05	8.56	3.81	2011-02-18	8.57	8.37	0.015	0.056	>50	2017-09-19	
4-RA-104-06	8.56	3.81	2011-02-18	7.98	7.91	0.037	0.020	>50	2017-09-19	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-07	8.56	3.81	2011-02-18	7.78	7.61	0.053	0.047	>50	2017-09-19	
4-RA-104-08	8.56	3.81	2011-02-18	8.18	8.00	0.050	0.050	>50	2017-09-19	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT Level II	2014-09-19	_____
Prepared	Trevor Nolan	UT technician, SNT-TC-1A UT Level I	2014-09-19	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2014-09-19	_____